

Lab 2: Memory

In this lab, you will participate in several memory demonstrations and record your results. Some questions may also ask you to predict and/or explain the results with respect to the concepts we discussed in the memory lecture. Your credit for this lab does not depend on your scores on any of

these tasks – the grade is based only on completion and the clarity of your answers.

Levels of Processing Experiment

1) What was the independent and dependent variable in the Levels of Processing experiment?

- The independent variable is the type of processing participants use when they are encoding the words. It could be shallow, intermediate or deep processing. The dependent variable is the number of words correctly recalled or recognized later, which shows how well the information was remembered.

2) Given the levels of processing model, what list would you predict that people would recall more words in? Why?

- Engaging with information on a deeper level will boost long term memory. So I would predict that people would remember more words from tasks that made them think harder, compared to those with simpler recall. This is likely because understanding the meaning of words creates a stronger memory link than just noticing how they look or sound, which helps in recalling the material later on.

3) How many words did you recall from List 1? 8

4) How many words did you recall from List 2? 7

5) Do your results confirm or refute the hypothesis above?

- My results support the hypothesis weakly because I remembered one more word from List 1 that involved deeper processing. This matches the prediction that deeper, more meaningful encoding leads to better memory performance.

Sperling's Iconic Memory Test

6) What is the difference between the whole report and partial report condition?

- The difference between the whole report and partial report condition was that in the whole report condition, participants try to recall all the letters they saw from the display. In the partial report condition, participants only recalled the letters from a specific row that plays right after the display disappears. The partial report usually produces a higher accuracy because people briefly retain more items in their visual sensory memory than they can verbally report before the image fades.

7) How many items did you remember on average in the whole report condition? 15

8) How many items did you remember on average in the partial report condition? 13

9) Given the number of items you remembered in the partial report condition, estimate how many items (minimum) were held in your iconic memory? How did you come up with this Estimate?

- I estimated around 10 based on the previous experiments and tasks.

Working Memory Capacity

10) Go to: <http://cognitivefun.net/test/7> and complete the digit span task. After you have gotten two trials correct you can click on the little up arrow (beside the number of digits) to increase the set size (i.e., to add another digit). The set size will begin at 3. Continue to increase the set size until you get 2 consecutive trials wrong, once you have gotten two consecutive trials incorrect then your digit span is the last set size that you were able to do correctly (or your current set size minus 1).

What was your digit span? 7

11) Go to: https://www.psychtoolkit.org/experiment-library/experiment_corsi.html Alternatively you may download the CorsiSpan file off of CCLE and run it offline. Both tasks are identical. You will watch a series of taps on the screen and you must use your mouse to repeat the sequence of taps after the computer tells you to “Go”. This task is easier to do if you have the sound on your computer turned on. At the end of the task, it will tell you your Corsi Span.

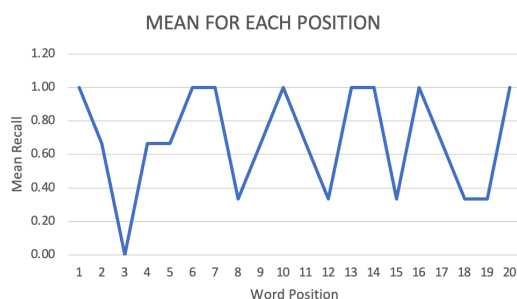
What was your Corsi span? 7

12) What component of the Baddley and Hitch (1974) model of working memory would you predict is involved in a digit span task? What component would you predict is involved in the Corsi span task?

- The digit span task mainly uses the phonological loop in Baddeley and Hitch’s working memory model. This part of our memory is all about holding onto and practicing verbal info, like numbers, by repeating them silently in our minds. On the other hand, the Corsi span task uses the visual spatial sketchpad, which deals with visual and spatial details. Because we had to remember where the blocks were and the order they were tapped, it checked how well we could picture spatial sequences, not just remember words or numbers.

The Serial Position Curve

13) Copy paste your serial position curve from your Excel/google sheets below.



14) What would happen to primacy and recency effects if you had to do something else in between hearing the end of the list and recalling the words? (e.g., I asked you to count backwards from 30).

- The recency effect would decrease because doing another task would prevent me from holding the last words in short term memory. The primacy effect would remain because the first few words were transferred into long term memory through practice.

Final Question

15) In preparation for your lab next week, we want you to brainstorm a few independent variables that think might influence people's ability to recall words in a free recall task. You could propose to manipulate some aspect of the stimuli (e.g., the types of words that participants memorize) or the method (e.g., the speed that words are shown, the font). You could also manipulate the instructions given to participants (i.e., asking participants to use different strategies to memorize the words). You will eventually design a free recall experiment based on one of the variables that your group comes up with.

- Manipulate: Words will appear on screen for either 1 second or 5 seconds each. Slower presentation gives more time for practice and memory, likely improving recall.